Health Canada Santé Canada

Leveraging the Gains: The WebDAIS-NESSTAR Project

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WebDAIS Project

- Initiated in January 2001
- Building on and leveraging the DDI, and the DDMS/DAIS and NESSTAR technologies
- Strategic approach:
  - Don't create competing servers - let's consolidate and strengthen the DDI standard to promote wider buy-in, especially by data producers and the official statistics community
    - The field of dreams - all data created to standards, fully integrated and immediately accessible. The 'data web'.
  - Capture and build on Europe's $5M (Can.) investment
  - Produce immediate results and payoffs for end users

Outline

- Why a standards-based approach
- From data graveyards to knowledge greenhouses
  - knowledge is what it's all about
- Goal of the WebDAIS Project
  - distributed capture and access
- NESSTAR - Open information space enabled by DDI
- Key thrusts for WebDAIS/NESSTAR
- What WebDAIS will look like
- Everyone can build on and use - support the standards and join in!.

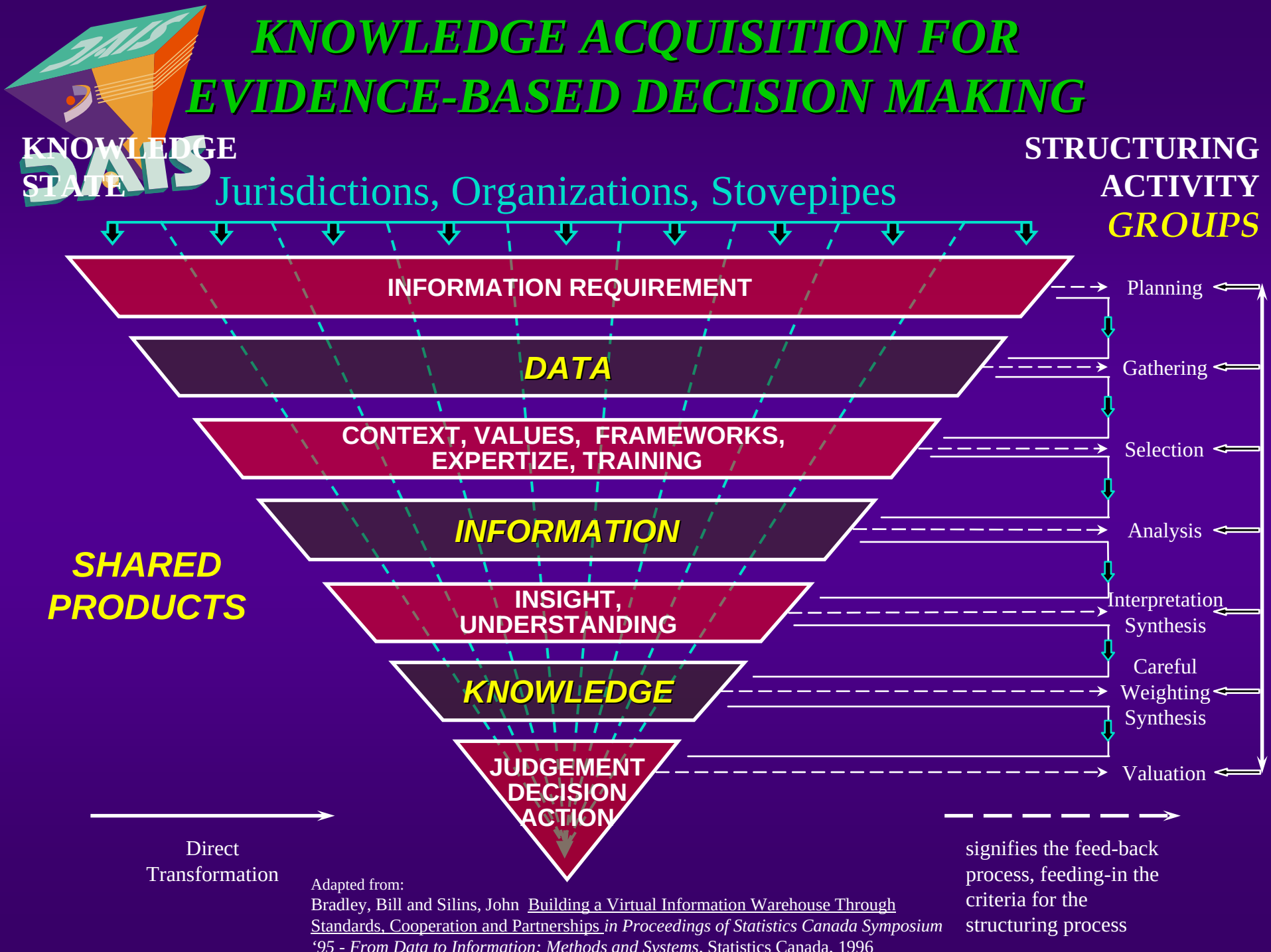
Why a standards-based approach

- Metadata standards
  - Make data immediately accessible in many formats and processing packages
  - Make data portable and exchangeable
  - Enable rapid access for decision support
  - Eliminate duplication of effort in documenting data for end user access
  - Enable users and support personnel to focus on data needs, substance and knowledge, not technical issues
  - Enable comparability, analytical integration across data sets, stovepipes and jurisdictions
  - Knowledge creation - the black gold of the new information age.
  - The way to sell data archiving, preservation

Data Graveyards to Greenhouses

- If you want to sell data (.. *archiving, preservation, documentation, sharing, national archives and funding, …*) relate it to *knowledge*.
- No one cares about data in its own right. Knowledge is what it is all about.
- In fact, it can be argued that even knowledge has little value until it is used to make a decision that enhances economic wealth or personal well being
- The DAIS functional model is based on a model of how knowledge is acquired and used to make 'evidence based decisions'

KNOWLEDGE ACQUISITION FOR EVIDENCE-BASED DECISION MAKING



Adapted from:
Bradley, Bill and Silins, John Building a Virtual Information Warehouse Through Standards, Cooperation and Partnerships *in Proceedings of Statistics Canada Symposium '95 - From Data to Information: Methods and Systems*, Statistics Canada, 1996

Metadata Standards (DDI) enable full implementation of the Knowledge Acquisition Pyramid

- Desktop access, with rapid movement in all directions through the pyramid
  - Horizontal integration across resources from many different organizations, jurisdictions, producers
    - Searches for variables across data sets.
    - Group comparable variables, measures, concepts
    - Create time series, indicators, comparative analyses
    - Automated table and chart packages
  - Vertical integration of data, information, knowledge
    - *Drill-up* from a variable to the analyses, research reports, tables in which it is used
    - *Drill-down* from research reports and tables to the

Metadata Standards (DDI) enable full implementation of the Knowledge Acquisition Pyramid

- Well demonstrated by Health Canada's DDMS and DAIS systems
  - DDMS - system for entering, editing, importing and exporting metadata in a standardized format
    - Used from the mid '80's to prepare and deliver standardized machine readable and hardcopy codebooks . Users have included the Canadian General Social Survey (GSS), Census, post censal survey programs, Canadian Association of Research Libraries and private sector polling firms.
  - DAIS - end user client for providing integrated access to microdata and associated tables and reports
      - A standard tool on all 7,000 desktops in Health Canada
- But … DDMS/DAIS are Health Canada only -

Goal of the WebDAIS Project

- To create a distributed version of DDMS/DAIS that will work across many autonomous local servers to better serve Health Canada and its partners at all levels of the health system.
  - 'Thousands of decisions are made throughout the health sector every day. Our goal is to ensure that they are made on the basis of the best available evidence.' - Denis Gauthier, ADM Health Canada
    - Relevant data, information and knowledge are created everywhere
    - Challenge is to capture and deliver them, when and where created or needed.
  - Need a highly distributed approach, with lot's of local autonomy for dealing with ownership, access, sharing/control issues. Security

DAIS and NESSTAR - Points of departure

- DAIS might be described as a closed, well-defined and centralised information space. All the resources are stored on one server and handled by a single publishing authority. This makes it possible to provide a complete and detailed resource catalogue (browse-list) and to keep this map updated whenever new resources are added to the system
- NESSTAR on the other hand is designed and implemented for the Internet. It is designed to support an unlimited and fully distributed information space where no single publishing authority has a complete overview of available resources.

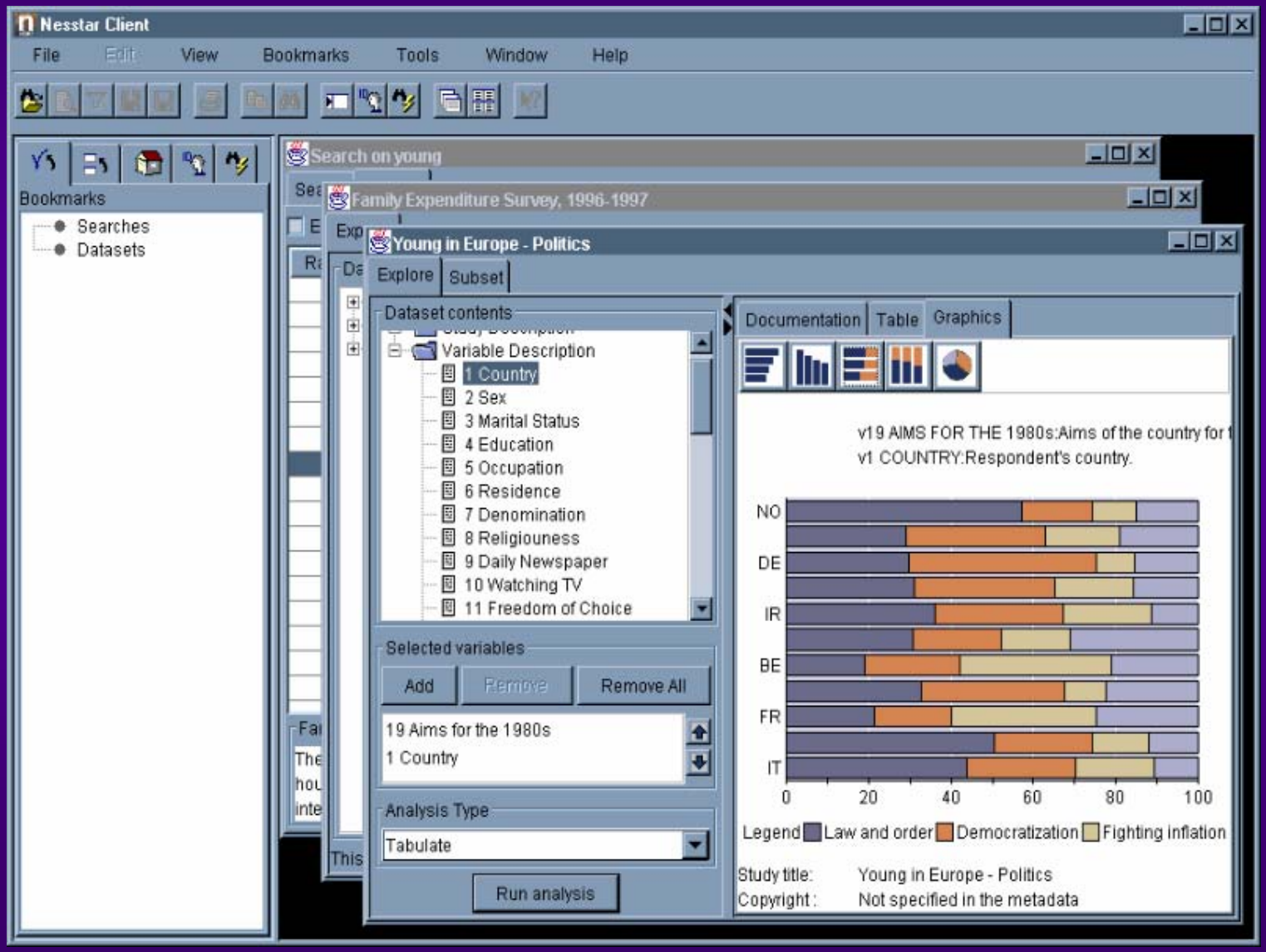
NESSTAR - An open information space based on DDI

- Works on basis of the dataset as an object
- Background in the distributed catalogue
- Finds and delivers a dataset with accompanying metadata for immediate browsing and downlaoding
- Facilitates creation of flexible hyperlinked information space

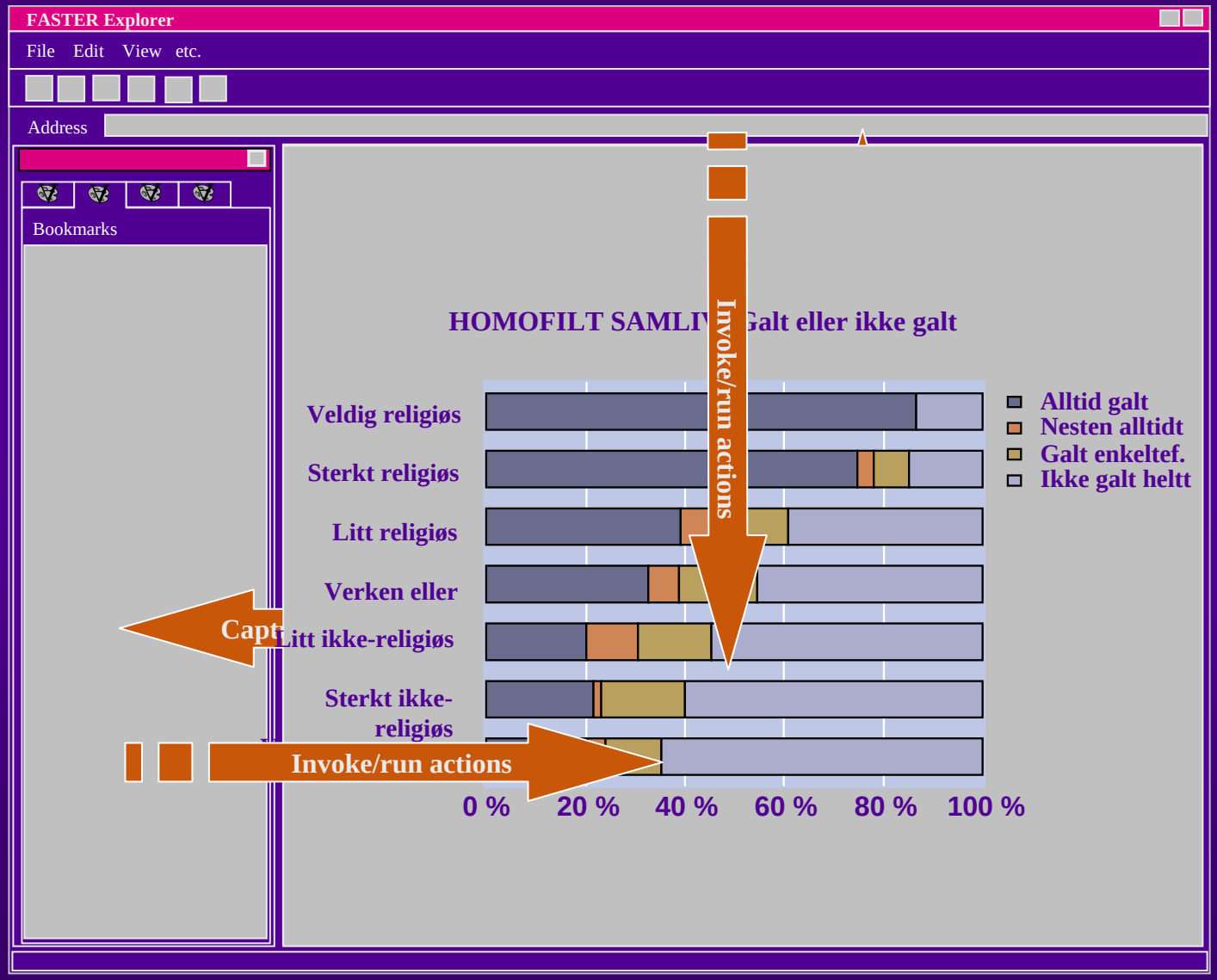
...ways of navigating an information space

- **Searching**: using terms, keywords and conditions to locate resources, either by searching the content of the resources directly or more indirectly by searching their corresponding metadata (Google style)
- **Browsing**: finding and navigating resources through catalogues, lists, directories etc. (Yahoo style)
- **Linking**: jumping from one resource to the next through embedded hyperlinks (Web-style). Although catalogues or directories usually will be driven by hyperlinks, there is a basic difference between the top down view of a browse-list and the network view of a hyperlinked information space.

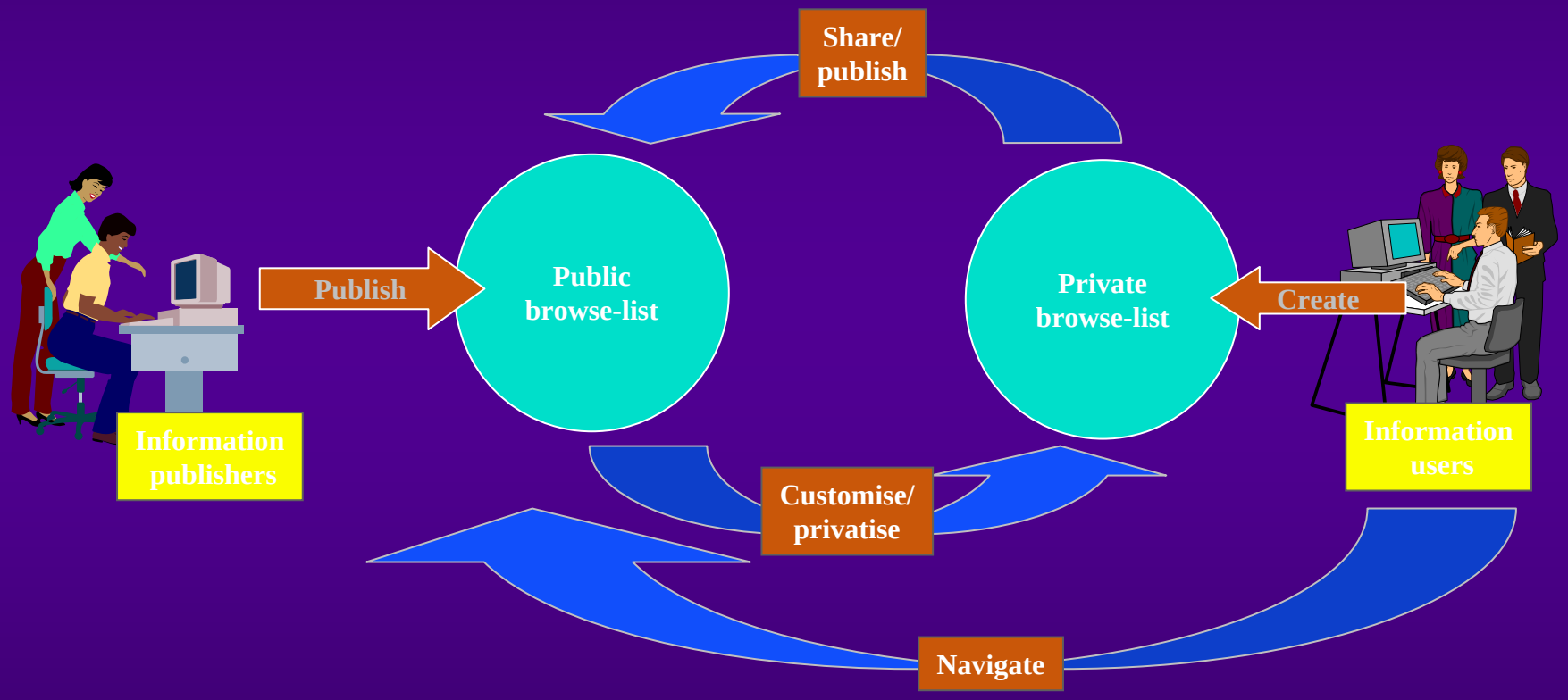
Private bookmarks in NESSTAR



...the nesstar bookmark/hyperlink system



Public versus private bookmarks



WebDAIS Project - capture the best of both systems

- NESSTAR brings:
  - Distributed server model built with RDF
  - Fully indexed text searching across multiple servers
  - High speed statistical engine
  - Flexible Java client
  - Powerful bookmark 'language'
- DDMS/DAIS brings:
  - Significant experience in metadata creation ('publishing') at source
  - Know-how in delivering to end-users in public sector environment
  - Integrated tables and knowledge products
  - Cross data set 'codebooks' (groups) for creating time-series, comparative analyses, automated report generation capabilities
  - Relational data base technologies

WebDAIS - What it will look like

- Single window of access for *all* information objects - data, information, knowledge products
- Shared across many local servers, organizations, jurisdictions, access and control policies, laws and practices …. *'information spaces'*
- Integrated across information objects *(drill-up/drill-down)* and across information spaces, *(drill sideways)* ie. through data, information knowledge and across organizations, jurisdictions, archives, users, producers …
- Let users choose and work at the level of the

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Dais
- Factsheets
  - Private Sector and Polls
  - Canadian Institute for Health Information/ICIS
  - Social & Health Reports/Rapports sur
  - National Population Health Survey
  - Special Topics/Sujets spéciaux
  - Target Groups/Groupes cibles
  - Diseases/Maladies
  - Health and General Surveys
  - The Census/Le Recensement - 1996
  - National Health Expenditures
  - Survey Questionnaires
  - About Dais - Workshop Slides and Background Papers
- Tables
- Datasets

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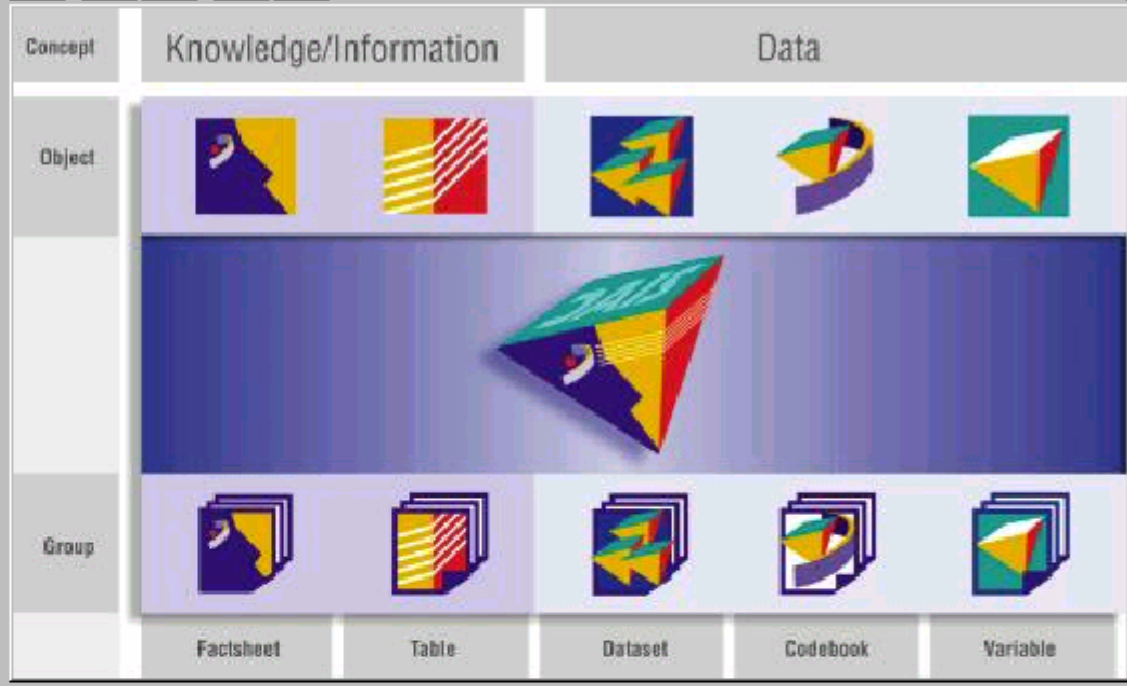

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- Canadian Tobacco Use Monitoring Survey
  - CTUMS99 - Wave 1: Fact Sheets
    - CTUMS99: FACT SHEETS
    - SUMMARY OF RESULTS
    - DESCRIPTION OF SURVEY
    - TRENDS IN SMOKING
      - CTUMS99:Trends in Smoking Be
      - NATIONAL POPULATION HEALTH
        - Codebook
      - SURVEY ON SMOKING IN CANAD
      - YOUTH SMOKING SURVEY, 1994
      - GENERAL SOCIAL SURVEY, 199
      - HEALTH PROMOTION SURVEY,
      - NATIONAL ALCOHOL AND DRUG
      - SURVEY OF SMOKING HABITS, 1
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      - CTUMS: Smoking Status, age, se
        - SMOKING STAT, AVG NO. OF
          - CANADIAN TOBACCO US
            - Codebook
        - SMOKING STAT, AVG NO. OF

Display pie chart

Health Canada Santé Canada

CTUMS CANADIAN TOBACCO USE MONITORING SURVEY

Wave 1 February-June 1999

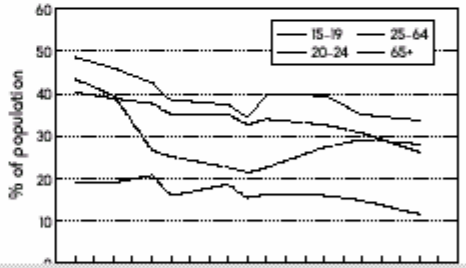
3. Trends in Smoking

Good News and Bad News

Smokers now constitute a smaller proportion of the Canadian population aged 15 and over (25%) than in either 1996 (29%) or 1994 (31%). This decline in smoking prevalence over the past few years has been similar for both males and females. However, it has not been equal across age groups, being confined mainly to Canadians aged 25 and older. Figure 3-1 summarizes smoking prevalence for key age groups from various Canadian surveys since 1981.[a]

The smoking rate for teens aged 15-19 has remained steady at 28% since 1996 and is almost the same level as in 1994 (27%). However, it remains significantly above

Figure 3-1
Current smokers, by age, age 15+, Canada, 1981–1999



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Description | Source | Author | Access | Note

Topics Include:

1. Good News and Bad News
2. Provincial Progress Highly Variable
3. Daily Consumption Declining
4. Access to Cigarettes

Figures:

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- Smoking
  - Canadian Tobacco Use Monitoring Survey
    - CTUMS99 - Wave 1: Fact Sheets
      - CTUMS99: FACT SHEETS
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      - DESCRIPTION OF SURVEY
      - TRENDS IN SMOKING
        - CTUMS99:Trends in Smoking Be
        - NATIONAL POPULATION HEALTI
          - Codebook
            - SECTION A1: DEMOGRAF
            - SECTION A2: GEOGRAPH
            - SECTION A3: EDUCATION
            - SECTION B: INCOME VAR
            - SECTION C1: LABOUR FC
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            - SECTION D7: HEALTH CA

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- OBJECTIVES
- SURVEY CONTENT
- SAMPLE DESIGN
  - 1996-97 SAMPLE DES
  - 1994-95 SAMPLE DES
- DATA COLLECTION
- DATA PROCESSING
- DATA QUALITY
- SURVEY ERRORS
- GUIDELINES - TABULATIC
- SAMPLE WEIGHTING GUI
- GUIDELINES - STATISTIC.
- RELEASE GUIDELINES
- APPROX. SAMPLING VAR
- WEIGHTING
- FILE USEAGE
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Identification | Description | Dimension | Methods | Access | Backgroun

Summary:

This is a Public Use Microdata File and contains microdata for all respondents.

The 1996-97 NPHS is the second wave of a continuing longitudinal survey of the health of Canadians. The survey follows and returns to the selected household respondent every

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- SUMMARY OF RESULTS
- DESCRIPTION OF SURVEY
- TRENDS IN SMOKING
  - CTUMS99:Trends in Smoking Be
    - AGE OF RESPONDENT (GRC
    - AVG CIGARETTES SMOKED F
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    - PROVINCE OF SELECTED RI
    - WHERE DO YOU USUALLY G
    - SEX OF RESPONDENT
    - WEIGHT VARIABLE: SIX MONT
    - AGE OF RESPONDENT
    - SEX OF RESPONDENT
    - TYPE OF SMOKER
    - HOW MANY CIGARETTES DO
    - FINAL WEIGHT
    - AGE GROUP
    - DERIVED PROVINCE

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AGE OF RESPONDENT (GROUPED)
WEIGHT VARIABLE: SIX MONTH WEIGHT
AVG CIGARETTES SMOKED PER DAY (DERIVED)
SEX OF RESPONDENT

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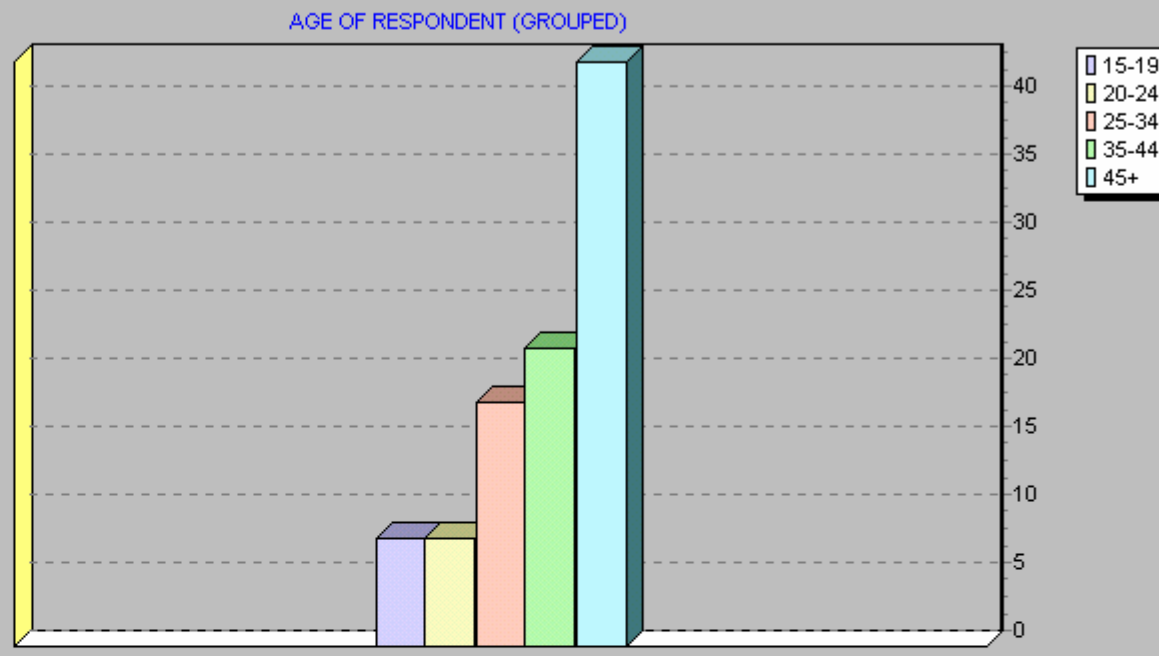

Summary | Source | Coverage | Comments | Interviewer | Weight

Age of respondent (grouped)

Response	Value	M	Weighted	% Excl	% Incl	Sample
15-19 year olds	1		2049311	8.5	8.5	3123
20-24 year olds	2		2045407	8.5	8.5	2551
25-34 year olds	3		4448564	18.4	18.4	1176
35-44 year olds	4		5238971	21.7	21.7	1415
45+	5		10396720	43.0	43.0	2899
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Total:			24178973	100%	100%	11164

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- DESCRIPTION OF SURVEY
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    - SEX OF RESPONDENT
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    - SEX OF RESPONDENT
    - TYPE OF SMOKER
    - HOW MANY CIGARETTES DO
    - FINAL WEIGHT
    - AGE GROUP
    - DERIVED PROVINCE

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AGE OF RESPONDENT (GROUPED)
WEIGHT VARIABLE: SIX MONTH WEIGHT
AVG CIGARETTES SMOKED PER DAY (DERIVED)
SEX OF RESPONDENT

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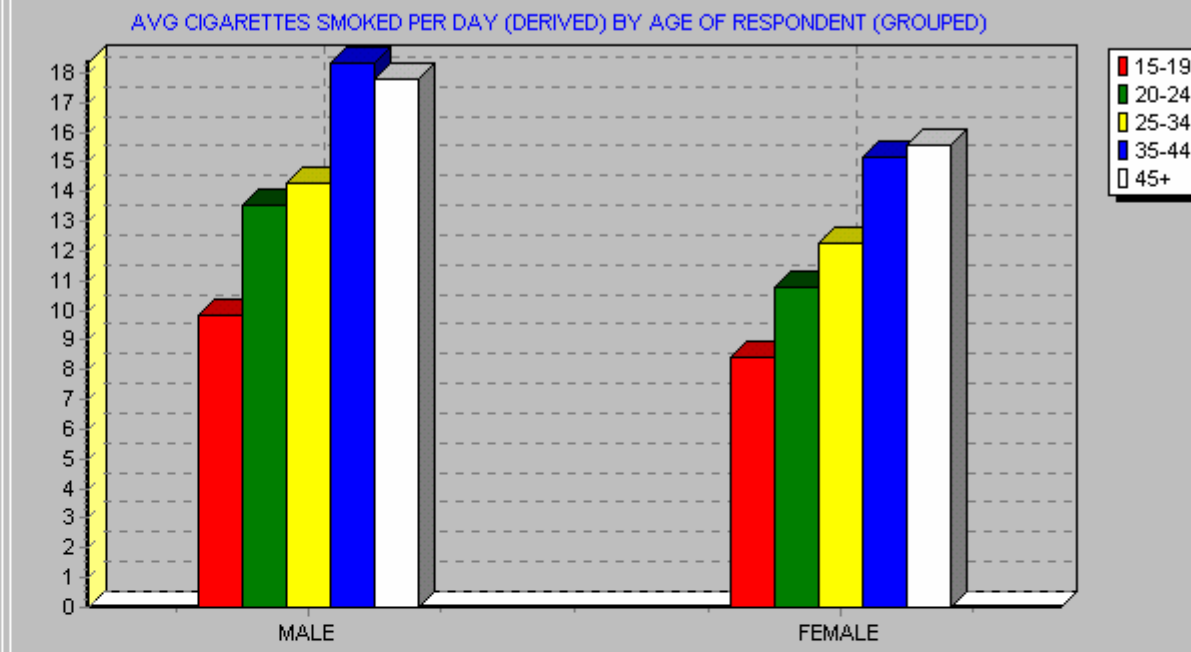

		G CIGARETTES SMOKED PER DAY (DERIVI			
RESPONDENT (GF	F RESPON	Mean	Min	Max	Valid N
15-19	MALE	9.81	0.00	40.00	472.00
	FEMALE	8.45	0.00	50.00	453.00
20-24	MALE	13.57	0.00	75.00	482.00
	FEMALE	10.75	0.00	75.00	469.00
25-34	MALE	14.30	0.00	54.00	177.00
	FEMALE	12.28	0.00	30.00	186.00
35-44	MALE	18.33	0.00	75.00	209.00
	FEMALE	15.15	0.00	50.00	246.00
45+	MALE	17.75	0.00	54.00	269.00
	FEMALE	15.54	0.00	50.00	293.00

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New | Favorites | Search

- SUMMARY OF RESULTS
- DESCRIPTION OF SURVEY
- TRENDS IN SMOKING
  - CTUMS99:Trends in Smoking Be

Concept | Knowledge/Information | Data

Object | Variable

Extract Wizard - Perform Extract

DAIS Extract Wizard

Package: SAS Windows

Dictionary: CANADIAN TOBACCO USE MONITORING'99 (W1 - PERSONS)

Case:

```
IF (
 (PROV = 24)
 OR
 (DVSST1 = 1)
 );
```

Output Location: C:\Daisnew\data\Extract\

☑ Run SAS Windows on extract

☐ Print

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ables | Factsheets | Tables

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AGE OF RESPONDENT (GRO
WEIGHT VARIABLE: SIX MONT
AVG CIGARETTES SMOKED F
SEX OF RESPONDENT
TYPE OF SMOKER 1 (DERIVE
TYPE OF SMOKER 2 (DERIVED)
PROVINCE OF SELECTED RESPONDENT
WHERE DO YOU USUALLY GET CIGARETTES
WEIGHT VARIABLE: SIX MONTH WEIGHT
AGE OF RESPONDENT

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- SUMMARY OF RESULTS
- DESCRIPTION OF SURVEY
- TRENDS IN SMOKING
  - CTUMS99:Trends in Smoking Be
    - AGE OF RESPONDENT (GRC
    - AVG CIGARETTES SMOKED F
    - TYPE OF SMOKER 1 (DERIVE
    - TYPE OF SMOKER 2 (DERIVE
      - Factsheets
        - General Reports - Car
        - Rapports - Tendances
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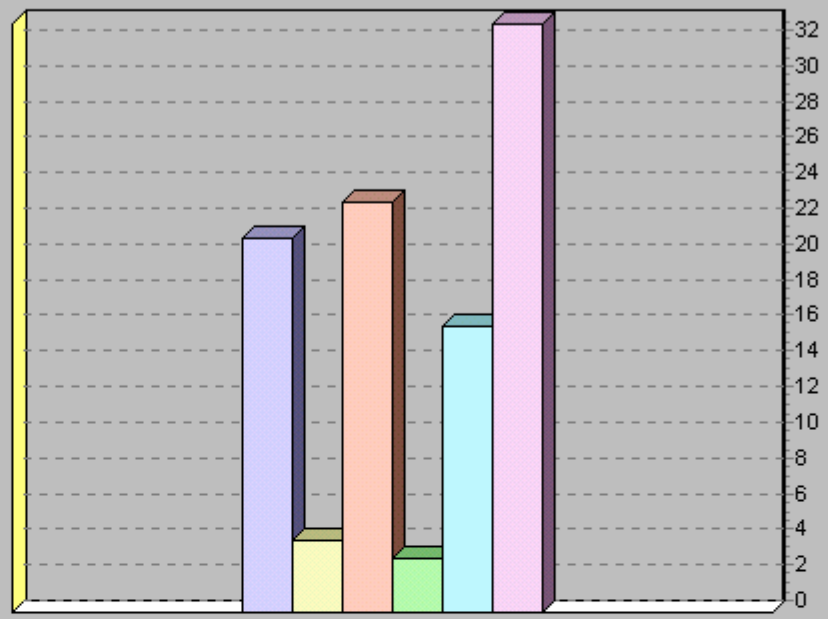

Summary | Source | Coverage | Comments | Interviewer | Weight

```
NOTE: New derived variable.
Smoking status - detailed - using traditional definition.
1 Current daily smoker = Q010=1
2 Current occasional smoker = Q010=2
3 Former daily smoker = Q010=3 and Q040=1 and Q060=1
4 Former occasional smoker = Q010=3 and Q040=1 and Q060=3
5 Experimental smoker = Q030=3 and Q040=3 and Q050=1
6 Lifetime abstainer = Q030=3 and Q040=3 and Q050=3
9 Not stated
History: Survey of Smoking in Canada
```

NUM 6:04:

Edit Group Browse Search User Window Help

(Variable) - TYPE OF SMOKER 2 (DERIVED)

Summary | Source | Coverage | Weight | Comments | Interviewer

Derived variable for type 2 smoking status.

Response	Value	Weighted	%Excl	%Incl	Sample
Current smoker	1	4965331	20.6	20.5	2545
Current occasional smoker	2	1033562	4.3	4.3	670
Former daily smoker	3	5529602	23.0	22.9	1846
Former occasional smoker	4	751899	3.1	3.1	236
Past experimenter	5	3879819	16.1	16.0	1804
Lifetime abstainer	6	7901369	32.8	32.7	3939
Not stated	99 M	117390	--	0.5	124
Total:		24178972	100%	100%	11164

Dataset | Codebook | Variable Groups | Tables | Factsheets

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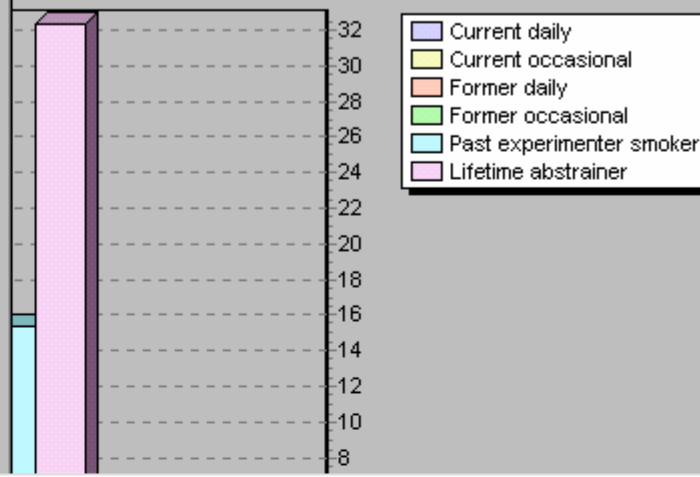

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- SUMMARY OF RESULTS
- SOMMAIRE DES RÉSULTATS
- TRENDS IN SMOKING
- PRÉVALENCE DE L'USAGE DE LA CIGARETTE, 1985-1999
- SURVEY ESTIMATES
- OVERALL CONCLUSIONS
- COMPARABILITÉ DES ENQUÊTES : MÉTHODOLOGIE ET QUEST
- FAITS SAILLANTS ET VUE D'ENSEMBLE
- PRÉSENTATION DES ESTIMATIONS D'ENQUÊTE
- CONCLUSIONS GÉNÉRALES
- KEEPING TRACK
- AU FIL DE L'ACTUALITÉ

SUMMARY OF RESULTS

Source:
CANADIAN TOBACCO USE MONITORING SURVEY

View | Dataset | Variable Groups | Tables

ables | Factsheets | Tables

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Summary

NOTE: New deriv
Smoking status
1 Current daily
2 Current occas
3 Former daily
4 Former occas
5 Experimental
6 Lifetime abst
9 Not stated
History: Survey

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Some Key Thrusts and Issues

- Metadata publishing - the DDMS replacement
- Take up by health partners and data producers
- Integration of DDI with ISO 11179
- Tables DTD and aggregate data methodology.
- The virtual information space - common browse lists and views across servers and organizations
- Security

Everyone can use

- Health Canada doesn't have a mandate to disseminate and support the technology for other users, but our partners in the project will have the rights to do so.
- Built upon international standards
- The critical issue is creating the metadata - the new data publisher will be key
- Designed for widespread distribution and integration
- Support the standards and join in!!